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Profit vs. Protection:
A Risk-Adjusted Comparison of Traditional and Renewable Energy Sectors

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1. Introduction

For the past two decades, the energy market was constantly redefined – and it seemed that we had finally found an answer to the impending climate crisis – green energy. Of course, this new sector had taken the economy by storm. There was uncertainty and volatility, and investors wanted to know one thing – was this market actually profitable? This essay attempts to investigate this dynamic landscape and how recent macroeconomic volatility over the last five years has impacted the risk-adjusted returns of two different hypothetical portfolios: one focused on green energy ETFs, and one focused on traditional fossil fuel ETFs.

By comparing a portfolio of renewable energy leaders against traditional energy majors, we aim to see if the green sector has successfully capitalized on the transition to clean power, specifically within the United States. We will analyze price data to determine if the volatility is worth the risk and decide whether or not the green sector is worth investing in.

This question is especially interesting to me as I have a deep interest in clean, renewable energy. I believe that it is the future of a sustainable earth. However, it is obvious that investors may have a different outlook on the benefits of clean energy. Does the market agree that it is worth investing in as a long-term growth option? And, more importantly, does the profit of oil and gas outweigh the environmental sacrifices we make to our earth?

2. Data Collection

To collect the daily price data, we utilize [Nasdaq.com](https://www.nasdaq.com) and focus on ETFs. ETFs offer a greater range of companies and can provide a more standardized outlook on the green and traditional energy sectors. We are going to look at data from January 1st, 2020 to January 1st, 2025. During this period, there have been several macroeconomic events in oil-based countries that may affect how the energy sector performs.

To begin, let us define our theoretical portfolios.

Portfolio A → Green Energy

- ICLN → iShares Global Clean Energy ETF
 - This security represents the global clean energy sector.
- TAN → Invesco Solar ETF
 - This security has broad exposure to solar energy volatility.
- ACES → ALPS Clean Energy ETF
 - This security has North American clean energy in its mix.
- QCLN → First Trust NASDAQ Clean Edge Green Energy ETF
 - This security focuses on clean energy technologies, with significant exposure to electric vehicles (EVs), batteries, and advanced materials.

Portfolio B → Fossil Fuels / Traditional Energy

- XLE → Energy Select Sector SPDR Fund
 - This security tracks S&P 500 energy companies (e.g. Exxon, Chevron, etc.)
- VDE → Vanguard Energy ETF
 - This security tracks broad exposure to US energy stocks.
- IXC → iShares Global Energy ETF
 - This security tracks globally integrated oil and gas.
- USO → United States Oil
 - This security tracks United States Oil supply.

For benchmarking, we are going to use SPY, representing the S&P500.

To analyze this data, we are going to utilize return versus volatility, Sharpe ratios, efficient frontiers, and correlation matrices. We hope to realize which sector has a better economic outlook, and use evidence to back the claim.

3. Analysis and Techniques Used

3.1 Returns Versus Volatility, Utilizing Sharpe Ratios

The goal of comparing returns and volatility was to find out if there was any correlation between the two. I hypothesized that **Portfolio B** would have higher returns for less volatility, and the opposite would be true for **Portfolio A**.

To begin this technique, I downloaded daily price data for both portfolios. I then calculated returns for both portfolios, using the following formula:

$$\frac{\text{today's return} - \text{tomorrow's return}}{\text{tomorrow's return}}$$

Then, using techniques learned during the portfolio playground module, I calculated average daily return using and daily standard deviation. We can then calculate the annualized average return by multiplying average daily return by 252, and the annualized volatility by multiplying standard deviation by the square root of 252.

Finally, we calculated the Sharpe Ratio. To do this, we assumed a risk-free rate of 3% which was subtracted from the annualized return, and divided by the annualized volatility. This gave us the risk-adjusted return of each ETF.

Then, in a table, the values of the Sharpe Ratio and Annualized Return were plotted to a scatter plot. SPY is indicated with a red marker as a benchmark reference. The Green Sector ETFs have green markers, the Traditional Sector have blue markers. The numbers attached to each ticker is the annualized return.



Figure 1. Risk-Adjusted Return versus Volatility of the Green Sector, Traditional Energy Sector

An initial analysis of this graph shows that my hypothesis was correct: the traditional sector generated greater returns with a high Sharpe Ratio. This means that for every unit of risk taken, these funds delivered significant excess returns.

However, at the bottom left, where the green sector ETFs are, we find that the Sharpe Ratios are between -0.1 and -0.3; these assets failed to compensate investors for the risk taken. SPY sits at a Sharpe Ratio of roughly 0.6 and a return of 13.5%. The traditional energy sector actually achieved a similar or slightly better Sharpe Ratio than the overall market, but with much higher absolute returns.

Usually, investors accept lower Sharpe Ratios in exchange for the potential of massive growth; however, we find that the Green cluster has negative returns paired with negative efficiency. It would not be advised to invest in the green sector, if we were looking purely at Sharpe Ratios.

3.2 Portfolio Optimization, The Efficient Frontier

To start, I hypothesized that the most efficient portfolio would not be made up of only traditional energy ETFs, but rather, combine both green assets and traditional assets. This is to ensure that the portfolio is not too heavily swayed one way, we must diversify our holdings as a safety measure against too much risk.

To generate the efficient frontier, we needed to utilize the solver function in Excel. First, I made a table with arbitrary weights for each of the ETFs. This is what we need to optimize. Then, I calculated the average return of each ETF. It is then critical to calculate the following values as a total portfolio value. The portfolio return was calculated using:

$$= \text{SUMPRODUCT}(\text{weights of ETFs}, \text{average return of ETFs})$$

The portfolio variance was calculated by multiplying the annualized covariance matrix (regular covariance matrix of all ETFs multiplied by 252) by the weights. It is imperative that the sum of the weights of the ETFs add up to 1 – if they go over 1, we have an impossible portfolio. The portfolio standard deviation is calculated by taking the square root of the portfolio variance. Then, we change the annualized return that we want for the portfolio to calculate the efficient frontier graph.

I chose to increment the annualized return by 2% starting at 0%. This is because the highest return of any of the ETFs is around 26%, so we cannot get a portfolio that has an annualized return of more than that.

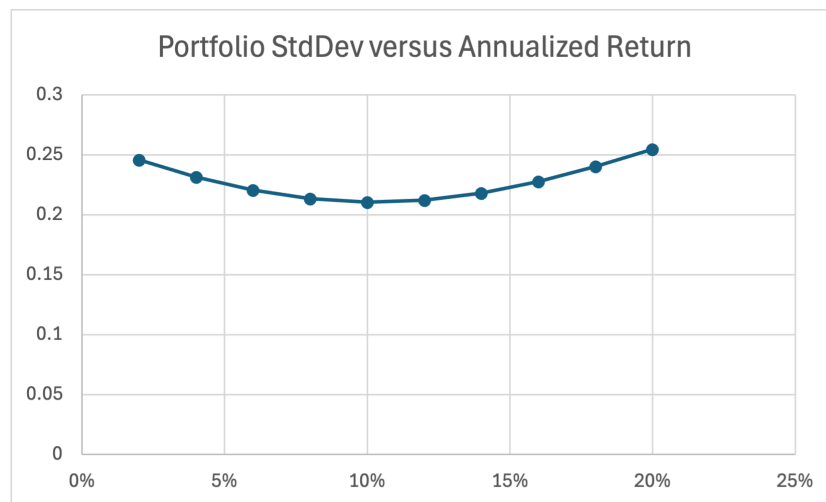


Figure 2. Portfolio Standard Deviation versus Annualized Return for a Mixed Optimized Portfolio

The minimum variance portfolio occurs at the lowest point in the graph – at 10% annualized return and a 21% standard deviation. This represents the safest possible portfolio we can construct; we cannot lower our risk below 21% no matter how we mix the ETFs. In this portfolio, the weights are distributed as follows:

ETF	ICLN	TAN	ACES	QCLN	XLE	VDE	IXC	USO
Weight	0.406258852	0	0	0	0	0	0.469678467	0.124062681

As a reminder, ICLN is a green ETF, and IXC and USO are both traditional energy ETFs. This proves the hypothesis and shows that despite the volatility of the green sector, there is still benefit in diversifying your portfolio. You cannot just hold traditional energy – you must blend them with uncorrelated assets. This model uses ICLN to dilute the high volatility of the traditional energy sector. It is important to note that ICLN is the broadest, most diversified global clean energy fund, making it the most safe green sector ETF to invest in. ISC has the lowest variance of any ETF at 0.060, and USO is a commodities fund, which behaves differently than equities, adding another layer of diversification.

As we move to the left side of the curve (0% to 10%), we observe the curve is inefficient. As you move left, your returns decrease but your risk increases. Given that the data shows that the green energy sector had negative returns and higher volatility, this left side represents portfolios heavily weighted towards green energy. No rational investor would choose a portfolio on the left. They essentially would be taking more risk for less money.

The right side of the right side of the curve (10% to 20%) is the true efficient frontier. As you move right, you gain higher returns, but you must accept higher volatility. This side represents portfolios more heavily weighted towards traditional energy, which had 20%+ returns in the data.

3.3 Correlation Matrix

So far, we have discovered that although Green Energy has terrible individual performances (negative Sharpe Ratios), it is most beneficial to include at least some Green Energy ETFs in your portfolio for a minimized standard deviation. A correlation matrix will explain why the math decides to keep a “losing” green asset.

I hypothesize that we will likely find traditional energy has a low correlation with green energy, thus proving that green energy can work as a diversifier, reducing the overall risk of the portfolio, even if its returns are lower during that specific five year window.

ETF	ICLN	TAN	ACES	QCLN	XLE	VDE	IXC	USO
ICLN	1	0.92536863	0.88748871	0.86083542	0.25611337	0.27088036	0.29344535	0.10636504
TAN	0.92536863	1	0.88781817	0.8661958	0.23666585	0.2495943	0.25720631	0.11976443
ACES	0.88748871	0.88781817	1	0.95656631	0.30662153	0.32393406	0.32186386	0.13961466
QCLN	0.86083542	0.8661958	0.95656631	1	0.31860745	0.33976061	0.3356972	0.1364703
XLE	0.25611337	0.23666585	0.30662153	0.31860745	1	0.99223219	0.97049717	0.65400591
VDE	0.27088036	0.2495943	0.32393406	0.33976061	0.99223219	1	0.97107262	0.66010115
IXC	0.29344535	0.25720631	0.32186386	0.3356972	0.97049717	0.97107262	1	0.66851182
USO	0.10636504	0.11976443	0.13961466	0.1364703	0.65400591	0.66010115	0.66851182	1

Figure 3. Correlation Matrix for Each ETF

Here, we can see that the hypothesis was correct. There is a distinct lack of correlation between the green sector and the traditional sector. For example, the correlation between ICLN (Green) and XLE (Fossil) is only 0.25. This is a

weak positive correlation, and means that when oil companies such as XLE move up or down, green energy companies such as those held in ICLN do not necessarily follow them. They are driven by different market forces – oil prices drive XLE, while interest rates and government policy drive ICLN.

This explains why the minimum variance portfolio in section 3.2 allocation about 40% to ICLN. Even though ICLN had negative returns, its negative correlation (0.25) acts as a shock absorber for the portfolio. If an investor only held XLE, the portfolio would go up and down completely in correlation with oil prices, which are not predictable at all. Adding ICLN smooths out the risk.

Assets that are in the same sector are highly correlated, meaning that similar assets are redundant. An optimized portfolio will pick the best in class ETFs and ignore the others. This validates why the optimization model zeroed out TAN, ACES, QCLN, XLE AND VDE – the model recognized them as “duplicates” of ICLN and IXC.

USO stands out as the most unique asset in this group. USO traces the price of oil futures, and behaves differently than XLE, which tracks the stocks of oil companies. USO’s correlation to ICLN is the lowest at 0.10, and its correlation to XLE is fairly low compared to others in its class at 0.65. This suggests that USO might be the single best diversifier in the list as it has almost zero relationship to green energy. This explains the 12% allocation to USO in the optimized portfolio.

4. Conclusions

This essay utilized techniques to determine if the societal promise of a green energy transition has translated into a better financial performance for investors over the volatile period of 2020 to 2025. By analyzing risk-adjusted returns with Sharpe Ratios, efficient frontiers and correlation matrices, the data delivers an interesting and nuanced verdict that challenges my original assumption that *good for the planet* automatically means *profitable for the portfolio*.

In terms of raw performance, the traditional energy sector was the undeniable winner. As detailed in Section 3.1, traditional energy ETFs capitalized on global supply constraints and macroeconomic shocks to deliver high Sharpe ratios and significant returns. In stark contrast, the Green Energy sector struggled, burdening investors with negative risk-adjusted returns and failing to compensate for its high volatility. If an investor were looking strictly at recent growth, the market’s answer is clear: the profitability of oil and gas currently outweighs the financial promise of renewables.

However, a purely profit-driven rejection of Green Energy would be a mathematical error. The Portfolio Optimization and Correlation analyses (Sections 3.2 and 3.3) reveal that the true value of the green sector lies not in growth, but in protection. The construction of the Minimum Variance Portfolio, which allocated roughly 40% to ICLN, proves that a portfolio consisting solely of high-performing traditional energy is inefficiently risky. Because the drivers of renewable energy prices (policy, interest rates) are largely uncorrelated with the commodity cycles driving oil majors, the green sector acts as a critical hedge.

Returning to the personal reflection in the introduction, while I believe green energy is the future of a sustainable earth, the market treats it, for now, as an insurance policy rather than a growth engine. The data suggests that we do not have to choose strictly between economic gain and environmental support; rather, the most "efficient" path forward requires integrating both. Rational investing demands that we hold green assets not just for their potential future dominance, but for the stability they provide today. Thus, even if the market does not yet fully reward the profitability of the transition, it undeniably validates the necessity of it.